ACADEMIC CAPSTONE & HACKATHON 2026

FileMind

AI-Powered PDF Vault & Knowledge Retrieval Engine

"Your documents. Organized. Searchable. Intelligent."

Core Architectural Pillars:

⚡ **Multimodal Document AI** • 🔒 **Multi-Tenant RLS Vault** • 🔍 **Hybrid Semantic Retrieval** • 📦 **150-File Bulk Ingestion**

Project Team & Affiliation:

Presenter: [Student Name / Lead Presenter] | Team Members: [Team Member 2, Team Member 3]

Department: [Department of Computer Science & Engineering] | Institution: [College / University Name]

Evaluation Track: [College Major Project Evaluation 2026] & [Annual College Hackathon 2026]

PROBLEM STATEMENT

The Crisis of Unstructured PDF Management

Traditional storage systems treat PDFs as opaque black boxes, creating massive knowledge fragmentation.

1. Opaque & Unsearchable Silos

Trapped in Local Folders

- PDFs are scattered across local folders, cloud drives, and downloads.
- Standard OS search only matches filenames, completely missing embedded body text and tables.
- Scanned invoices, contracts, and image-based PDFs remain 100% invisible to search.

2. Manual Reading Fatigue

Hours Lost Skimming Pages

- Extracting critical facts (e.g. invoice total, vendor name, contract clause) requires tedious manual opening.
- Users re-read the same lengthy reports repeatedly just to locate a single data point.
- Zero automatic categorization or structured schema generation.

3. Security & Scale Bottlenecks

Fragile Ad-Hoc Solutions

- Generic cloud drives provide storage but lack multi-tenant row-level access control for documents.
- Conventional desktop OCR tools choke on large batches and fail on modern complex layouts.
- No automated queue recovery, leading to frozen uploads and lost metadata.

PRODUCT INNOVATION

Our Solution: The FileMind AI Vault

Transforming passive PDF storage into an active, intelligent, and secure retrieval ecosystem.

❌ TRADITIONAL PDF STORAGE

Passive, dumb folders with manual maintenance

- **Manual Filename Search:** Only finds documents if you recall the exact file naming syntax.
- **Opaque Scanned Pages:** Cannot inspect image-based or multi-column PDFs without expensive desktop software.
- **Zero Metadata Extraction:** Invoice numbers, amounts, and tags must be manually typed into spreadsheets.
- **Single-File Friction:** Must upload and view files one by one; fails on large batch volumes.

✅ FILEMIND AI-POWERED VAULT

Autonomous document intelligence with semantic discovery

- **Multimodal Understanding:** Google Gemini extracts categories, tags, key-value data, and page-by-page text.
- **Semantic Hybrid Search:** Find concepts (e.g. 'quarterly cloud spending') even if the exact keyword never appears.
- **High-Capacity Concurrency:** Bulk upload up to 150 PDFs (49 MiB each) with 4-way upload and 2-way AI sliding windows.
- **Enterprise Security:** PostgreSQL Row-Level Security, advisory locks, and cryptographic fencing leases prevent data leaks.

VERIFIED CAPABILITIES

Core Implemented Features

Every feature shown is fully implemented, tested, and running in the FileMind codebase.

[BULK INGESTION]

150-File Bulk Upload

Sliding-window concurrency engine handling up to 150 files per batch, 49 MiB per file, with partial-failure isolation.

[MULTIMODAL AI]

Gemini Multimodal Analysis

Strict JSON schema extraction for document category, custom tags, and dynamic key-value metadata with injection defense.

[SEARCH ENGINE]

Hybrid Semantic Retrieval

Cosine similarity over vector embeddings combined with full-text search, featuring a 500-entry in-memory LRU cache.

[VAULT SECURITY]

Multi-Tenant Isolation

Supabase PostgreSQL Row-Level Security (RLS) guaranteeing strict per-user isolation across database records and storage.

[RESILIENCE]

Durable Queue & Leases

Atomic job claiming with SKIP LOCKED and UUID fencing tokens, recovering stale processing jobs automatically.

[OPTIMIZATION]

Lossless PDF Optimization

Structural PDF optimization via pdf-lib saving up to 15% storage while safely skipping signed or encrypted documents.

SYSTEM PIPELINE

End-to-End Ingestion & Processing Workflow

Step-by-step lifecycle from client-side drag-and-drop to vector indexing and viewer retrieval.

01

Client Batch Upload

User drops up to 150 PDFs. Frontend validates 49 MiB limit, requests signed upload URLs, and streams files directly to private storage.

02

Quota & DB Commit

Authoritative RPC verifies storage object size & ownership. Advisory lock checks 2 GB user storage quota & inserts pending document.

03

Queue Lease Claim

Background worker invokes claim RPC with SKIP LOCKED. Assigns a cryptographic UUID fencing token to prevent duplicate work.

04

Gemini Multimodal AI

PDF is processed via Gemini Files API. Model cascade (3.6-flash -> 3.8-flash -> preview) extracts structured metadata with backoff.

05

Atomic Finalization

Transaction commits category, tags, and pages while validating fencing token. Clears lease token and schedules vector indexing.

06

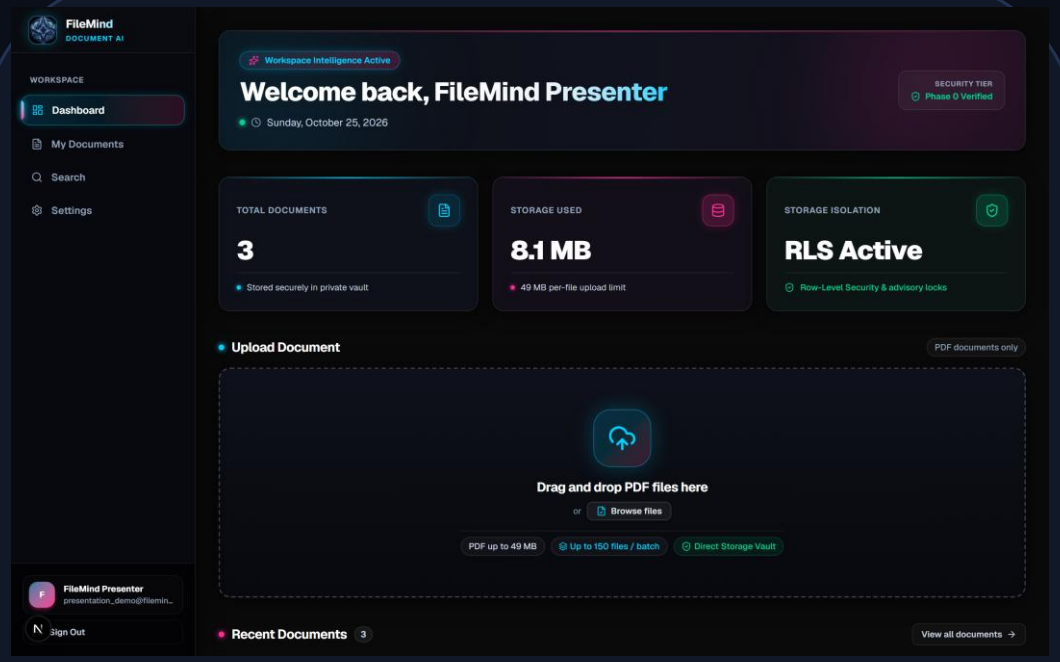
Instant Retrieval

Page chunks are indexed with 768-dim embeddings. User searches return ranked matches with snippet previews in <200ms.

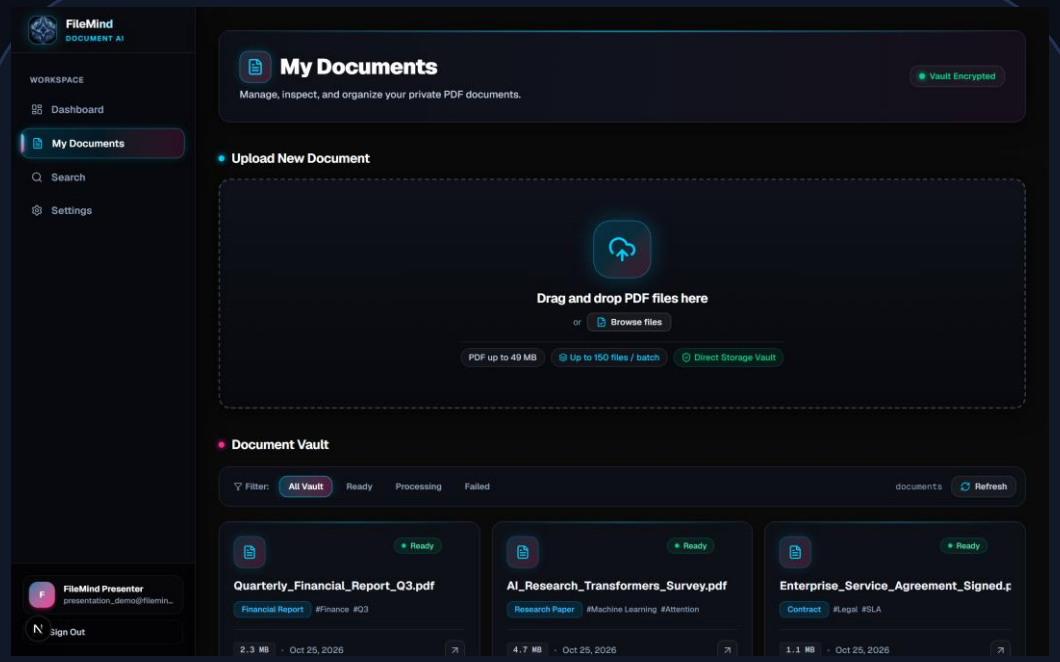
PRODUCTION UI

Live Application Screenshots

High-contrast dark-mode interface built with Next.js 16, Tailwind CSS v4, and Lucide icons.



Dashboard & Upload Dropzone:
Features real-time metric cards (Document Count, Storage Quota, Processing Rate), 150-file bulk drag-and-drop zone, and authoritative document status.



Documents Vault & Semantic Explorer:
Interactive repository displaying automated category badges (Financial Report, Contract, Research), AI-extracted semantic tags, file size savings, and actions.

ENGINEERING FOUNDATION

Verified Technology Stack

Engineered with modern, production-grade frameworks ensuring performance, scalability, and type safety.

FRONTEND FRAMEWORK

Next.js 16.3 + React 19

Next.js 16 with Turbopack for lightning-fast compilation, React 19 Server Components, and client-side streaming UI.

STYLING & DESIGN SYSTEM

Tailwind CSS v4 + Lucide

Modern CSS-first theme with zero runtime overhead, high-contrast accessible typography, and Lucide React icons.

DATABASE & SECURITY

Supabase (PostgreSQL 15+)

PostgreSQL with strict Row-Level Security (RLS), transactional advisory locks, and atomic PL/pgSQL stored procedures.

MULTIMODAL AI

Google Gemini API (@google/genai)

Direct multimodal PDF processing via Gemini Files API with structured JSON response schema enforcement.

SEARCH & EMBEDDINGS

Vector Embeddings + LRU Cache

768-dimensional document embeddings paired with an in-memory 500-capacity LRU cache for sub-200ms queries.

STORAGE & OPTIMIZATION

Supabase S3 Storage + pdf-lib

Private user-isolated storage buckets with lossless structural PDF stream optimization via pdf-lib.

SYSTEM DESIGN

Layered System Architecture

High-level architectural blueprint showing clear separation of concerns and secure communication boundaries.

TIER 1: PRESENTATION LAYER (CLIENT-SIDE)

- Next.js 16 Responsive UI: Dashboard, Document Vault, PDF Viewer, Semantic Search
- Client Batch Upload Manager: Sliding-window queue (max 4 concurrent uploads, per-file progress)

TIER 2: GATEWAY & SERVERLESS ROUTING (NEXT.JS 16)

- API Endpoints: /api/documents, /api/search, /api/cron/process-queue
- Execution Handlers: Next.js 16 after() background execution + Supabase Auth Session Validation

TIER 3: DATA & ATOMIC TRANSACTION TIER (SUPABASE / POSTGRESQL)

- Row-Level Security (RLS) enforcing tenant isolation; Private S3-Compatible Storage bucket 'documents'
- Stored Procedures: create_document_with_quota, acquire_document_processing, finalize_document_processing

TIER 4: INTELLIGENCE & ACCELERATION ENGINE

- Gemini Multimodal Files API: Structured category, metadata, and page text extraction
- Search Optimization: 500-LRU embedding cache, snippet coalescing, and lossless pdf-lib compression

COMPETITIVE VALUE

What Makes FileMind Different?

Direct feature comparison against traditional folder storage, cloud drives, and generic OCR utilities.

Capability / Feature	Local OS Folders	Generic Cloud Drive	FileMind AI Vault
Search Methodology	Exact filename match only	Keyword match on text PDFs	**Semantic & Concept Retrieval (768-dim embeddings)**
Multimodal Extraction	None (Manual reading)	Basic OCR (Text only)	**Structured JSON Schema (Category, Tags, Key-Values)**
Scanned / Image PDFs	Opaque & Inaccessible	Limited text layer export	**100% Understood via Multimodal Vision AI**
Bulk Upload Management	OS Copy (No validation)	Slow sequential sync	**150-file Batch Sliding Window (4x network concurrency)**
Security & Tenant Isolation	Local user permissions	Coarse folder sharing links	**Database Row-Level Security (RLS) & Fencing Leases**

TECHNICAL RIGOR

Engineering Challenges & Solutions

Real software engineering hurdles encountered during development and the verified solutions implemented.

1. Gemini 503 Spikes & Model Cascade

Challenge: During high-demand periods, Google's standard tier returns 503 UNAVAILABLE, causing queue processing failures.

Solution: Engineered an intelligent model cascade: gemini-3.6-flash -> gemini-3.8-flash -> gemini-3-flash-preview with exponential backoff and jitter.

2. Serverless Timeouts on 150-File Batches

Challenge: Processing 150 PDFs in a standard Next.js 60-second serverless execution budget leads to function timeout kills.

Solution: Implemented sliding-window concurrency (max 4 uploads, max 2 AI workers), Next.js after() execution, and a pg_cron background consumer.

3. Concurrency Race Conditions & Deadlocks

Challenge: Simultaneous file confirmations cause duplicate rows and quota circumvention under concurrent uploads.

Solution: Added transactional PostgreSQL advisory locks pg_advisory_xact_lock(user_id) and atomic SKIP LOCKED queue claiming with UUID fencing leases.

4. Safe PDF Optimization Without Corruption

Challenge: Compressing digitally signed or encrypted PDFs invalidates cryptographic signatures and renders files unreadable.

Solution: Conservative pre-checks detect /ByteRange signatures and /Encrypt dictionaries, safely skipping them while optimizing standard structural streams.

ROADMAP

Future Scope & Planned Advancements

Strategic enhancements designed to expand FileMind into an enterprise-ready knowledge platform.

PHASE 2 (NEXT MILESTONE) — Cross-Document Synthesis & Multi-PDF Chat

- Ask questions across your entire library: 'Compare revenue across all Q1, Q2, and Q3 financial reports.'
- Synthesize comparative summaries and highlight conflicting clauses across multiple contracts.

PHASE 3 — Interactive Visual Citations in PDF Viewer

- Exact bounding-box highlighting on the original PDF canvas showing the exact paragraph source for every AI answer.
- Clicking a search result immediately scrolls the viewer to the exact matched text coordinates.

PHASE 4 — Local / Air-Gapped LLM Execution

- Support on-premise inference with Ollama and vLLM for sensitive medical or classified defense documents.
- Complete zero-data-egress option allowing organizations to run FileMind fully offline.

PHASE 5 — Collaborative Team Vaults & RBAC

- Shared workspace vaults with granular role-based access control (Admin, Editor, Read-Only Reviewer).
- Audit trail logs tracking who viewed, exported, or modified document metadata.

PROJECT SUMMARY & DEMO

FileMind — The Intelligent Document Vault

"Stop searching for documents. Start finding answers with FileMind."

Key Project Takeaways:

✓ **Problem Solved:** Replaces passive, unsearchable folders with an active AI-powered document intelligence vault.

✓ **Production Quality:** Engineered with Next.js 16, Supabase PostgreSQL, and Google Gemini with verified test suites.

✓ **Resilient Concurrency:** Tested with 150-file bulk batches, automated recovery, and transactional advisory locks.

✓ **High Privacy:** Database-enforced Row-Level Security ensuring strict per-user vault isolation.

Ready for Evaluation & Live Demo:

1. Live Ingestion Walkthrough:

Drag-and-drop a batch of test invoices, papers, and legal agreements.

2. AI Extraction & Semantic Search:

Watch the Gemini pipeline extract categories and search by natural concepts in real-time.

Presenter Contacts & Resources:

GitHub: [github.com/your-username/filemind]

Email: [your-email@college.edu] | Demo URL: [http://localhost:3000]

Thank you! We welcome questions from our teacher and judges.